Special Topics: Machine Learning (ML) for Networking

COL867
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Synthetic Data Generation

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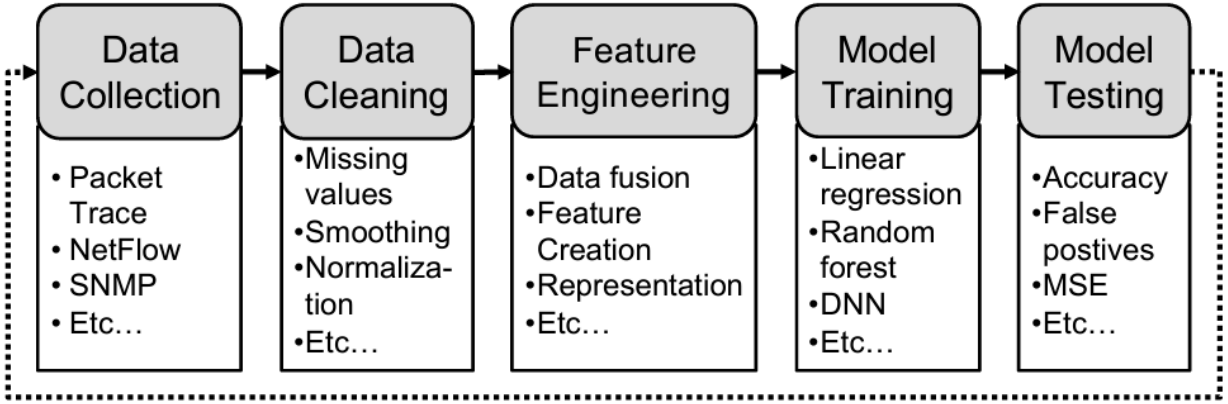
ML for Networks

Module 1: Case studies of specific network learning tasks

Module 2: Task-agnostic automatic ML pipelines for networks

Module 3: Beyond feature engineering and modeling

- Network telemetry
- Robustness
- Explainability
- Formal verification
- **Synthetic data generation**



Why?

① Reduce cost of acquiring data

② To handle underrepresented data issues → (Security → Attack data)

③ Privacy issues

Synthetic Data Generation

What: artificially create network data (e.g., packet traces) similar to real-world network behavior

Why:

- Privacy and security
- Simulation of rare events
- Data scarcity
- Controlled experiments

How?

1. Simulation ⟷ Emulation in controlled condition
   ↳ Building simulators can be challenging
2. Model-driven
   → $P(x|\theta)$
   ↳ what is the prob. distribution?
3. Use Machine learning

How to generate synthetic network data?

- Simulation (e.g., NS3)
- Model-based approach
- Machine learning-based approach → Deep learning generative model

Using Generative Models

$\sim p(x|\theta)$

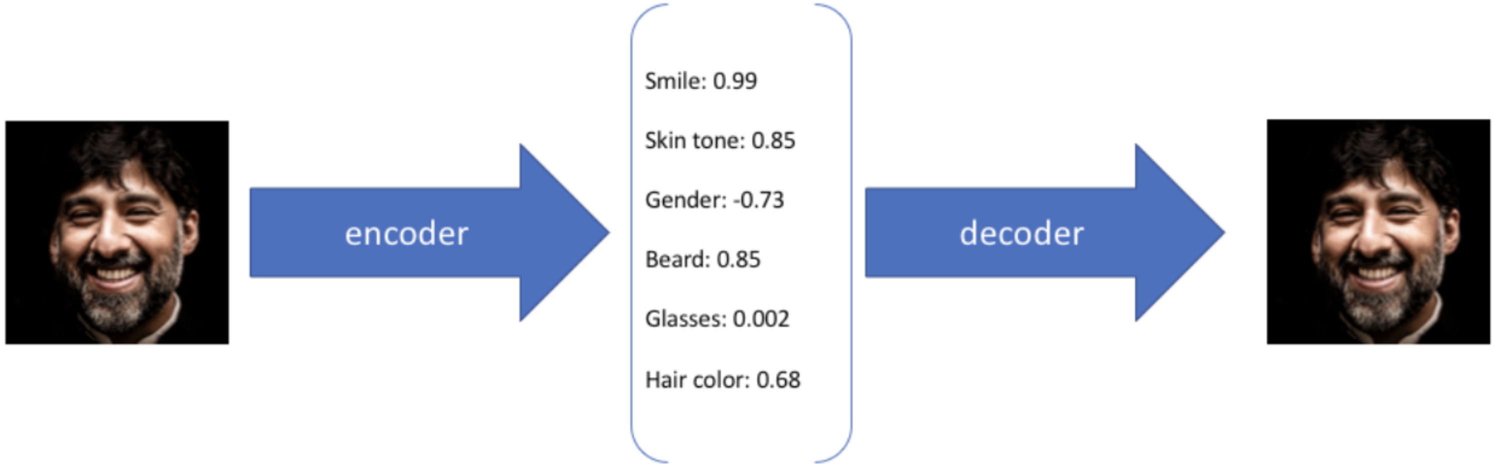
$\sim p(x, z|\theta)$

latent variable

- Generative models that assume the distribution depend on some latent variables z

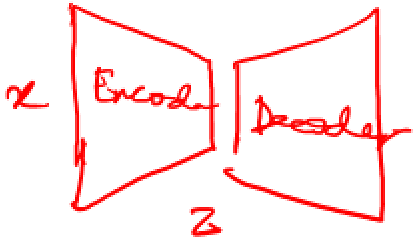
$p(x|z)$

$p(z)$



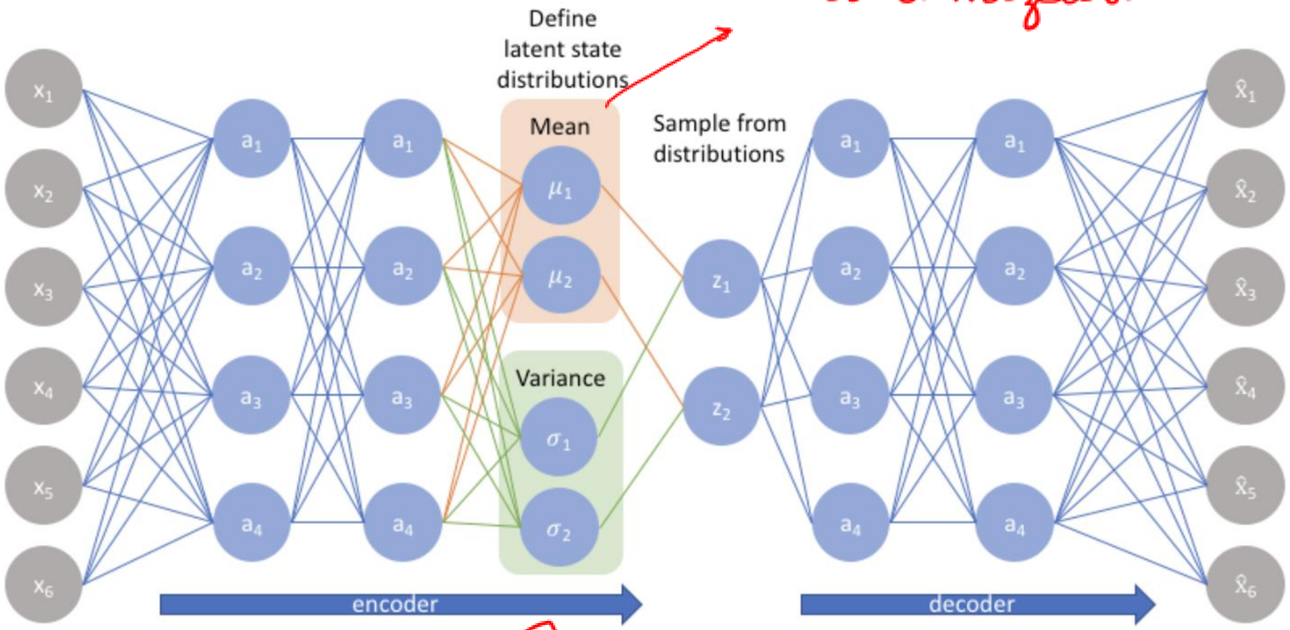
Latent attributes

Examples of Generative Models

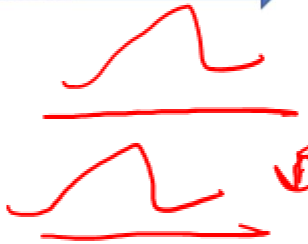


- Variational Autoencoder (VAEs)

Customization



Human Face →



Smile

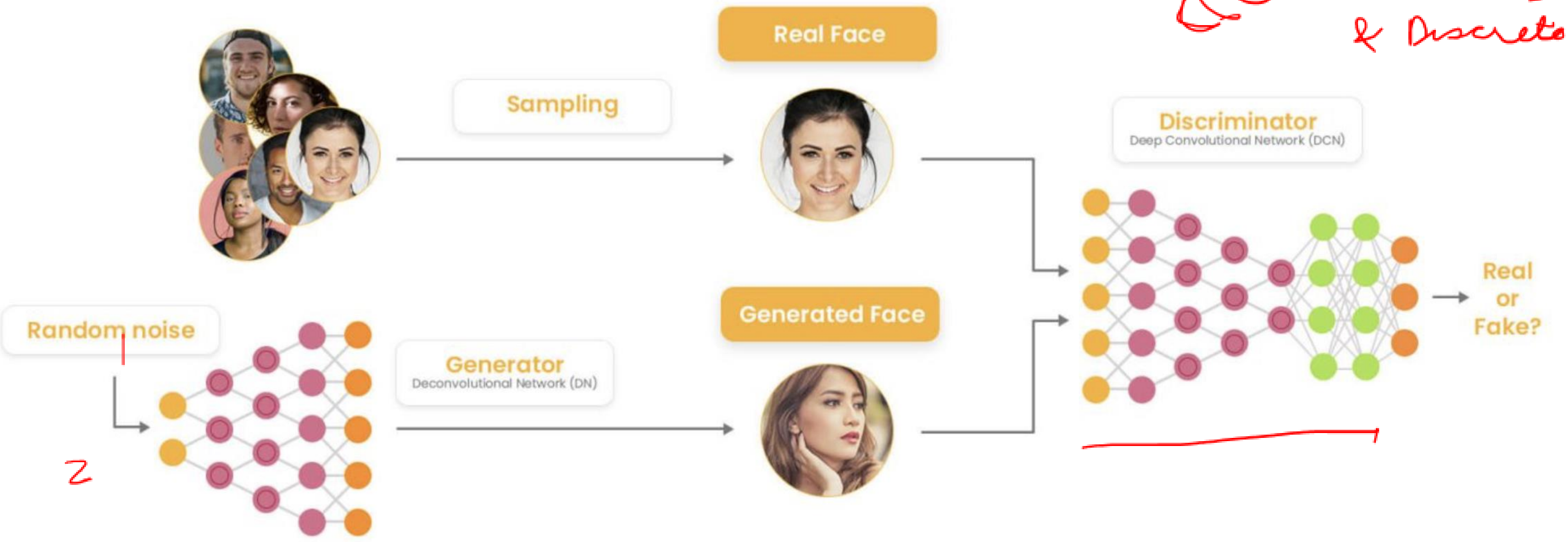
Skin color

Time series GAN → NetShare

Generative Adversarial Networks

① Time series data

② Continuous & Discrete



z

1 real

0 gener

Loss function

$$\min_G \left[\max_D \mathbb{E}_{\mathbf{y}\sim p_{\text{data}}}[\log D(\mathbf{y})] + \mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}}}[\log(1 - D(G(\mathbf{z})))] \right]$$

NetDiffusion

Diffusion model

NetDiffusion: Network Data Augmentation Through Protocol-Constrained Traffic Generation

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Datasets of labeled network traces are essential for a multitude of machine learning (ML) tasks in networking, yet their availability is hindered by privacy and maintenance concerns, such as data staleness. To overcome this limitation, synthetic network traces can often augment existing datasets. Unfortunately, current synthetic trace generation methods, which typically produce only aggregated flow statistics or a few selected packet attributes, do not always suffice, especially when model training relies on having features that are only available from packet traces. This shortfall manifests in both insufficient statistical resemblance to real traces and suboptimal performance on ML tasks when employed for data augmentation. In this paper, we apply diffusion models to generate high-resolution synthetic network traffic traces. We present *NetDiffusion*[1], a tool that uses a finely-tuned, controlled variant of a Stable Diffusion model to generate synthetic network traffic that is high fidelity and conforms to protocol specifications. Our evaluation demonstrates that packet captures generated from NetDiffusion can achieve higher statistical similarity to real data and improved ML model performance than current state-of-the-art approaches (e.g., GAN-based approaches). Furthermore, our synthetic traces are compatible with common network analysis tools and support a myriad of network tasks, suggesting that NetDiffusion can serve a broader spectrum of network analysis and testing tasks, extending beyond ML-centric applications.

CCS Concepts: • **Networks** → **Network simulations**; • **Computing methodologies** → **Neural networks**.

Additional Key Words and Phrases: Network traffic, synthesis, diffusion model

Diffusion Models: Intuition

The essential idea, inspired by non-equilibrium statistical physics, is to systematically and ***slowly destroy structure in a data distribution*** *through an* ***iterative forward diffusion process****. We then learn a* ***reverse diffusion process*** *that* ***restores structure in data****, yielding a highly flexible and tractable generative model of the data.*

Diffusion model to generate network traffic traces

nPrint → Bit vector

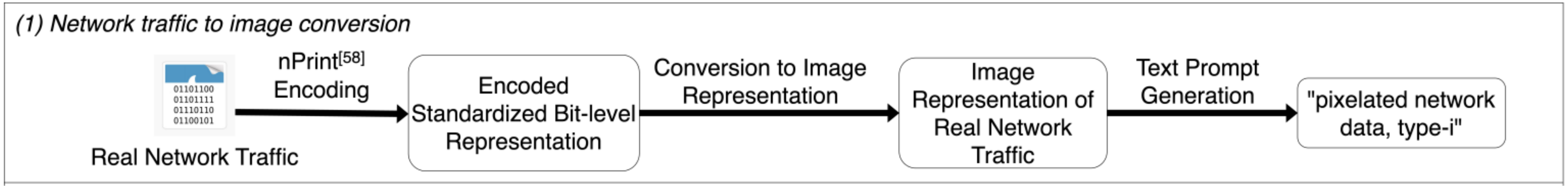
Demo

How do we use diffusion models for network data?

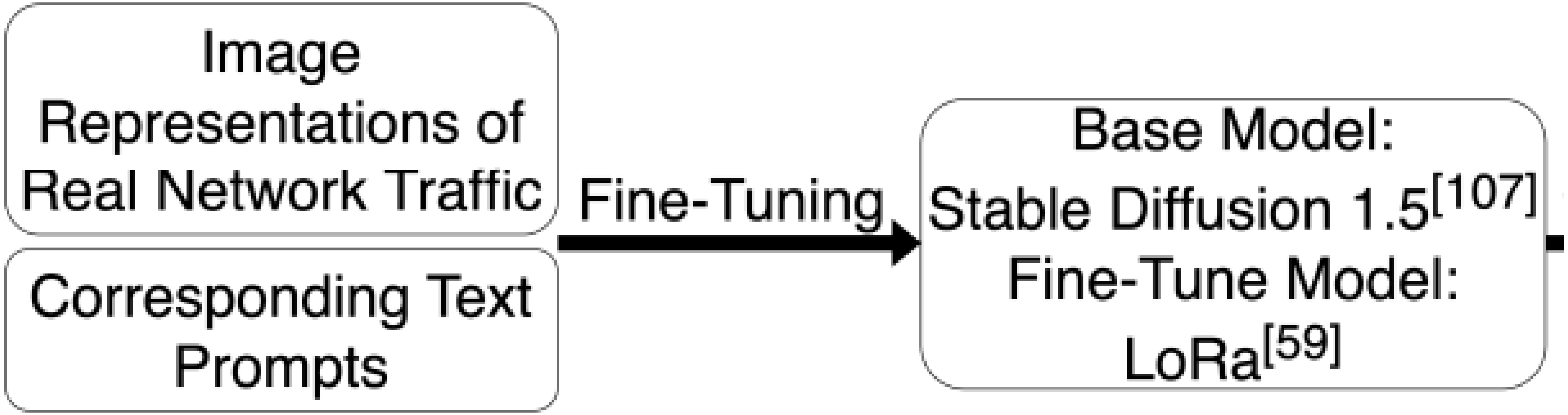
- Problem statement: Generate synthetic PCAPs

- **Challenge:** Diffusion model is mostly defined for image data

PCAP to Image Conversion



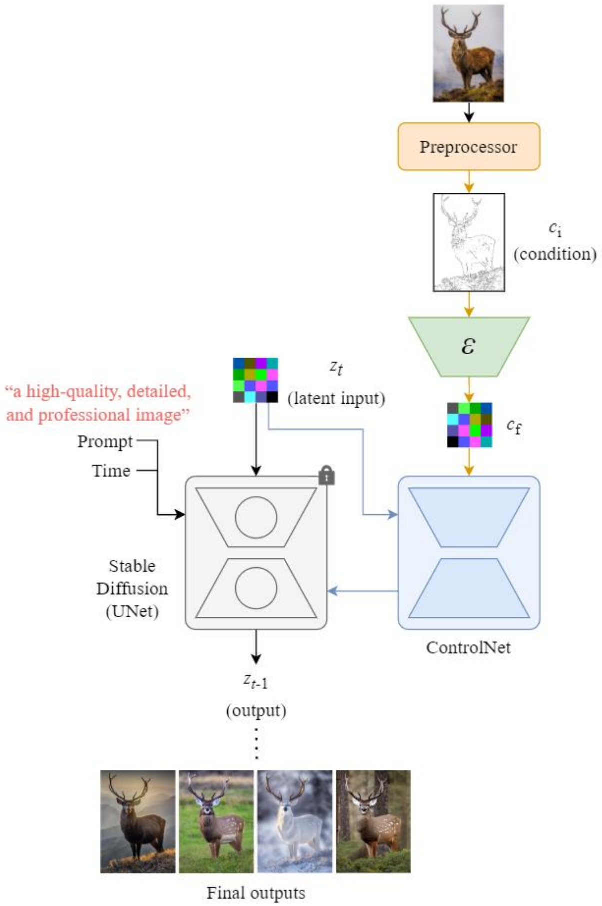
Fine-tune a pre-trained model



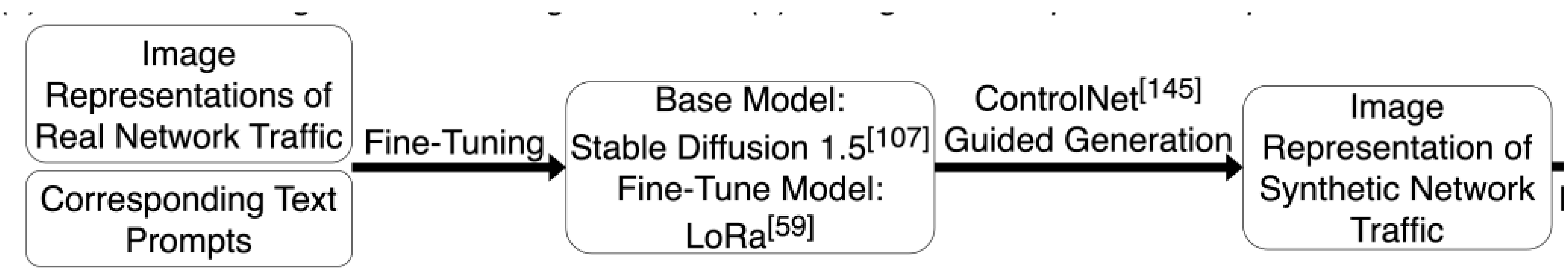
The generated images may have

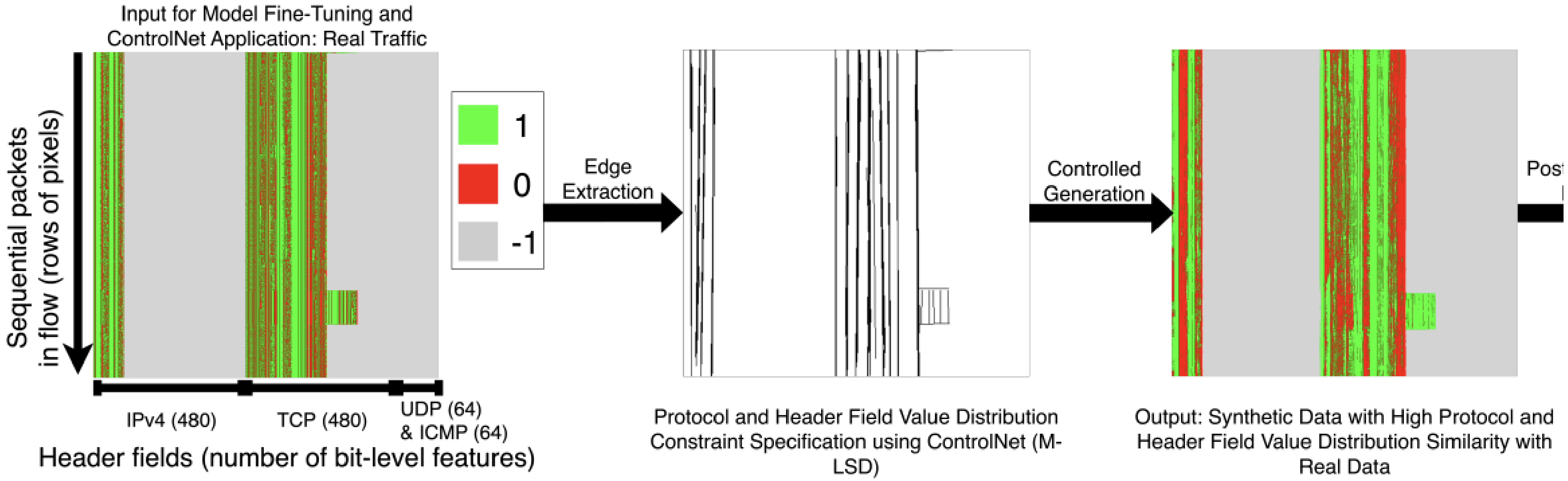
ControlNet

Additional neural networks that take additional hints to control the layout, shape, or structure of the output



NetDiffusion





Protocol Compliance Heuristic

- Define intra- and inter-packet rules

```
IPV4 Header
├── Internet Header Length (IHL, 4 bits: Specifies the header size in 4 byte words
│   (5–15 words). Calculation: Binary of ((number of present IPV4 header bits
│   (typically 160) + number of present IP option bits) / 32 bit-per-word); Default
│   (no IP options): 0101 (4 words).
├── Total Length (16 bits: Defines total packet bytes (20-65535 bytes).
│   Calculation: Binary of (TCP/ICMP/UDP header size (e.g, Data offset in TCP) + IHL
│   + TCP/ICMP/UDP payload size (if present)); Must be greater or equal to than
│   TCP/ICMP/UDP header size + IHL
├── Data (variable length = Total Length (bytes) – IHL*4(bytes/word))
│   ├── TCP Header (variable length, depends on Data Offset field)
│   │   ├── Data Offset (4 bits: Indicates TCP header size in 4-byte words (5–15
│   │   │   words). Calculation: Binary of ((number of present TCP header bits
│   │   │   (typically 160) + number of present TCP option bits) / 32 bit-per-word);
│   │   │   Default (no TCP options): 0101 (4 words).
│   │   └── Options (Variable size: (Data Offset-5)*4 bytes if Data Offset > 5, i.e.,
│   │       5 words = 20 bytes = min TCP header size; Max 320 bits in nPrint since
│   │       option can have a maximum size of 40 bytes (60 –20), using –1 to denote
│   │       missing ones)
│   └── TCP Payload (variable length: Total Length (bytes) – IHL*4 (bytes/word) –
│       Data Offset*4 bytes)
```

(a) Intra-packet.

```
TCP Session
├── Packet 1 (Src to Dst)
│   ├── Ports (consistent for all packets from Src); Seq Num (randomly initialized
│   │   for Packet 1 and 2)
│   └── Flags: No ACK for first packet with SYN set
├── Packet 2 (Dst to Src: SYN–ACK)
│   ├── Ports (consistent for all packets from Dst); ACK = Seq Num of Packet 1 + 1
│   └── Flags: SYN and ACK set to respond to SYN from Packet 1
├── Packet 3 (Src to Dst: ACK)
│   ├── Sequence = Sequence of Packet 1 + 1; ACK = Sequence of Packet 2 + 1
│   └── Flags: ACK set to respond to SYN from Packet 2
├── Subsequent Packets
│   ├── Src to Dst
│   │   ├── Seq Num (Seq Num of last packet sent by Src + data size (bytes) sent in
│   │   │   the last packet)
│   │   └── ACK (Seq Num of last packet received from Dst + data size (bytes)
│   │       received in the last packet)
│   ├── Dst to Src
│   │   ├── Seq Num (Seq Num of last packet sent by Dst + data size (bytes) sent in
│   │   │   the last packet)
│   │   └── ACK (Seq Num of last packet received from Src + data size (bytes)
│   │       received in the last packet)
│   └── Flags: ACK set in all packets after the initial exchange to acknowledge
│       receipt of data
├── Packet Loss Scenario:Packet N lost in transmission
│   ├── Packet N+1 (sent by the same host as Packet N, repeats ACK of Packet N–1)
│   └── Retransmission of Packet N (Seq Num and ACK same as lost Packet N)
└── others
```

Evaluating Synthetic Data Quality

Evaluating Synthetic Data Quality

Statistical similarity

Data Format	Generation Method	All Generated Features			Ex. Common Feature: Protocol		
		Avg. JSD	Avg. TVD	Avg. HD	Avg. JSD	Avg. TVD	Avg. HD
NetFlow	Random Generation	0.67	0.80	0.76	0.82	0.99	0.95
	NetShare	0.16	0.16	0.18	0.04	0.03	0.04
Network Traffic (pcap)	Random Generation	0.82	0.99	0.95	0.83	1.00	1.00
	NetDiffusion	**0.04**	**0.04**	**0.05**	**0.02**	**0.03**	**0.02**

Evaluating Synthetic Data

Training/Testing	Data Format (Generation Method)	Post-Gen. Heuristic	Highest Accuracy (Model) Macro-level	Micro-level
Real/Real	Network traffic (N/A)	N/A	1.00 (SVM)	0.978 (SVM)
	Netflow (N/A)	N/A	0.86 (RF)	0.648 (DT)
Synthetic/Real	**Network traffic (NetDiffusion)**	**Not Used**	**0.738 (DT)**	**0.262 (DT)**
		Used	**0.676(DT)**	**0.222 (DT)**
	Netflow (NetShare)	N/A	0.396 (RF)	0.140 (SVM)
Real/Synthetic	**Network traffic (NetDiffusion)**	**Not Used**	**0.542 (SVM)**	**0.249 (SVM)**
		Used	**0.529 (SVM)**	**0.182 (SVM)**
	Netflow (NetShare)	N/A	0.503(SVM)	0.102 (RF)

Evaluating Synthetic Data Quality

Training Data (Balancing Source)	Model	Test Accuracy Pre/Post Balancing	Δ Acc.
Network Traffic (NetDiffusion)	**RF**	**0.982 → 0.986**	**0.004** **Facebook (0.955 → 1.00)**
	DT	**0.973 → 0.982**	**0.009** **Meet (0.909 → 1.00)** **Zoom (0.955 → 1.00)**
	SVM	0.991 → 0.991	0
Netflow Data (NetShare)	RF	0.645 → 0.628	-0.017
	DT	0.603 → 0.600	-0.003
	SVM	0.290 → 0.290	0

Discussion

Next Class

- Project presentation in the next two classes

- Slots will be shared tomorrow